

PREDATORY INVITATIONS AS EXTENDED PHENOTYPE:

A Parasitic Spontaneous Order Framework for Academic Publishing

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Replication repository: <https://github.com/adrianlerer/predatory-invitations-as-extended-phenotype>

ABSTRACT

Predatory publishing is consistently framed as a problem of individual fraud: bad actors mimicking legitimate journals to extract fees from credulous academics. This paper argues that framing is wrong, and wrong in a theoretically tractable way. Drawing on Dawkins' (1982) Extended Phenotype Theory (EPT) and Hayek's (1973) model of spontaneous order, I introduce the concept of Parasitic Spontaneous Order (PSO): an emergent institutional niche in which the unsolicited invitation email is not incidental spam but a phenotypic extension of a replicator that has adapted, over roughly two decades of selection pressure, to exploit the very mechanisms that legitimate scholarship depends upon. I construct a corpus of eight documented invitation cases received between 2025 and early 2026, including one confirmed legitimate invitation used as control. Applying a five-criterion PSO Index (measuring Mismatch, Mimicry, Hook Architecture, Friction Minimization, and Automation Signatures), I find that seven of eight cases (87.5%) satisfy the full PSO threshold. Pattern analysis identifies six recurrent strategies, the most prevalent being credential mismatch signaling (7/7 predatory cases) and flattery personalization (6/7). The PSO lifecycle, reconstructed from corpus evidence, follows eight identifiable stages: SSRN/preprint scraping, mimetic invitation dispatch, response filtering, simulated peer review, payment extraction, credential vending, author network exploitation, and niche stabilization. Unlike principal-agent or regulatory capture models, the PSO framework predicts that no individual actor need intend parasitism: the behavior is an equilibrium, not a conspiracy. This has direct implications for institutional design, bibliometric policy, and the legal treatment of academic fraud. Replication materials are archived at the GitHub repository cited above.

Keywords: predatory publishing, extended phenotype, spontaneous order, parasitic institutions, academic publishing fraud, PSO Index, replicator dynamics

1. INTRODUCTION

Every academic researcher with a visible online presence receives them. The subject line varies; the structure does not. A journal with an impressive-sounding name and a website indistinguishable from that of a legitimate publisher invites the recipient to submit a manuscript, often praising a specific recent paper by name. The deadline is urgent. The review is rapid. The fee appears only after acceptance. And then the paper, if submitted and accepted, joins a bibliography that no serious indexing service records and that most hiring committees have learned, unevenly, to recognize as a problem.

The standard account of this phenomenon treats it as fraud: a deliberate deception perpetrated by bad actors, solvable in principle through better detection, blacklists, author education, and, where legal systems permit, criminal prosecution. Jeffrey Beall's lists, active from 2008 until their controversial removal in 2017, embodied this approach. The Cabell's Predatory Reports database, launched in 2017, extended it. Grudniewicz et al.'s 2019 consensus definition in *Nature* treated predatory journals as those that 'prioritize self-interest at the expense of scholarship' through false information and lack of transparency. Each of these contributions was valuable. None of them explains why the phenomenon persists, scales, and evolves faster than countermeasures.

This paper proposes a different analytical register. If predatory publishing were simply individual fraud, it would be unstable: fraud schemes collapse when detection costs fall below marginal gains for victims. Instead, predatory publishing has expanded. A 2021 analysis by Shen and Björk estimated over 420,000 articles published in potential predatory journals in 2014 alone, a figure that has grown since. The invitation infrastructure has become sophisticated enough to scrape institutional repositories, parse author metadata, and generate personalized flattery at scale. That is not the behavior of a fraud scheme. It is the behavior of a niche.

The theoretical framework I introduce, Parasitic Spontaneous Order (PSO), synthesizes two bodies of literature that have not previously been brought into direct contact. The first is Richard Dawkins' Extended Phenotype Theory (EPT), developed most fully in his 1982 monograph. EPT argues that the replicator, the unit of selection, can be understood as extending its phenotypic effects beyond the organism into the environment, including the behavior of other organisms. A beaver's dam, the nest architecture of a weaver bird, and the chemical manipulation of a host ant by a parasitic fungus are all, in the EPT account, phenotypic expressions of genetic replicators. The second body of literature is Friedrich Hayek's analysis of spontaneous order: the emergence of stable institutional patterns from decentralized individual actions without central design. What I argue is that the predatory invitation email is a phenotypic extension of a cultural replicator, the predatory publishing memplex, and that this replicator has found a stable ecological niche through a spontaneous order process operating on academic incentive structures.

This is not a metaphor. I operationalize both claims. The PSO Index provides a falsifiable instrument for classifying individual invitation emails along five dimensions that correspond to the structural requirements of the EPT-Hayek synthesis. The PSO lifecycle maps the stages through which a predatory operation achieves niche stability. The empirical corpus of eight cases, documented in the replication repository archived at GitHub (github.com/adrianlerer/predatory-invitations-as-extended-phenotype), provides the evidentiary foundation.

The contribution of this paper is threefold. Conceptually, it is, to my knowledge, the first application of Extended Phenotype Theory to predatory publishing, and the first synthesis of EPT with Hayekian spontaneous order in any domain. Empirically, it introduces a validated, replicable classification instrument. Practically, it explains why individual-fraud-focused countermeasures are insufficient and points toward institutional design responses that address the selective environment rather than the individual actor.

The paper proceeds as follows. Section 2 develops the theoretical framework in detail, covering EPT, spontaneous order theory, and the PSO synthesis. Section 3 describes the methodology, including corpus construction, the PSO Index, and its validation. Section 4 presents results: descriptive statistics, pattern analysis, and lifecycle reconstruction. Section 5 discusses implications, compares PSO to competing frameworks, and addresses limitations. Section 6 concludes.

2. THEORETICAL FRAMEWORK

2.1 Extended Phenotype Theory: The Replicator Beyond the Organism

Dawkins introduced the extended phenotype not as a rhetorical flourish but as a claim about the proper unit of evolutionary analysis. The selfish gene account, developed in the 1976 work of that title, placed the gene at the center of selection theory. The 1982 elaboration extended the implication: if genes are replicators, their phenotypic effects are whatever affects their replication frequency, and there is no principled reason to limit those effects to the body of the organism containing the gene.

The canonical examples from biology are illuminating for the institutional case. The fluke *Dicrocoelium dendriticum* manipulates the behavior of ants to cause them to climb blades of grass and wait to be eaten by sheep, completing the parasite's lifecycle. The cuckoo egg, by mimicking the coloring of its host's eggs, exploits the host's recognition heuristics to secure parental investment for a foreign replicator. The common thread: a replicator achieves transmission by altering the behavior of systems it does not contain, using those systems' own functional architecture against their inclusive fitness interests.

The application to cultural replicators, what Dawkins in 1976 called memes, is more contested and requires care. I do not invoke the meme concept in its popular, imprecise usage. I

use it in the technical sense developed by Dennett (1991, 1995): a cultural replicator that propagates by influencing the cognitive and behavioral apparatus of its hosts. A predatory publishing operation is, in this framework, a memplex: a cluster of coordinated replicators (business model, invitation template, fake editorial board structure, simulated review process) that propagate together because their joint effect on the selective environment exceeds the sum of their individual effects.

Two features of EPT are particularly relevant here. First, the replicator's phenotypic extension can include the behavior of organisms that are not its hosts. The cuckoo's egg does not 'contain' the host bird's genes; the fluke does not 'contain' the ant's. Similarly, the predatory publishing memplex does not 'contain' the academic author who responds to an invitation. It extends its phenotypic reach into the author's decision-making architecture by exploiting cognitive biases, institutional incentive structures, and informational asymmetries that the legitimate publishing system itself created. Second, EPT is agnostic about intentionality at the level of the replicator. Genes do not intend anything. The question is whether the replicator propagates, not whether anyone designed it to propagate. This is the feature that distinguishes the EPT account from fraud-focused analyses: EPT predicts persistence and adaptation without requiring conspiracy or even stable organizational continuity.

2.2 Spontaneous Order: Institutions Without Designers

Hayek's concept of spontaneous order, developed most systematically in *Law, Legislation and Liberty* (1973), describes the emergence of stable institutional patterns from the uncoordinated actions of individuals pursuing local goals. The price system is Hayek's paradigm case: no individual designs it, no central authority maintains it, yet it aggregates and transmits information that no central planner could access. The key mechanism is selection: practices that coordinate behavior successfully survive and diffuse; practices that fail disappear.

Hayek distinguished two types of order: taxis, deliberate constructions, and cosmos, spontaneous emergences. Predatory publishing is, in Hayekian terms, a cosmos: no single actor designed the elaborate mimicry infrastructure, the scraping methodology, the flattery templates, or the editorial board forgery conventions. These practices emerged through selection operating on variation in a competitive landscape where the resource is author fees and the threat is detection.

The spontaneous order account has a counterintuitive implication that is important for policy. Because the order is not designed, it cannot be dismantled by identifying and punishing designers. Beall's blacklists demonstrated this: each delisted journal spawned successor operations with marginally improved mimicry. The selective pressure exerted by detection and blacklisting is, from the PSO framework's perspective, simply an environmental variable that shapes the next generation of replicators. This is not pessimism about countermeasures; it is a specification of what kinds of countermeasures can work. I return to this in the Discussion.

2.3 Parasitic Spontaneous Order: The Synthesis

PSO names the specific configuration in which a spontaneous order achieves stability by exploiting the functional architecture of an adjacent legitimate order. Three conditions are necessary and jointly sufficient. First, the parasitic order must use mimicry: it must produce signals that the target order's detection mechanisms treat as evidence of legitimacy. Second, the exploitation must be structural rather than individual: the parasite exploits features of the host order that the host order cannot remove without damaging its own function. Third, the parasitic order must be self-reinforcing: each successful exploitation creates conditions that make subsequent exploitation more likely, generating the ecological niche stability that distinguishes a PSO from a one-off fraud.

In predatory publishing, these three conditions are satisfied as follows. Mimicry: predatory journals reproduce the visual, typographical, and structural signals of legitimate journals (ISSN numbers, editorial board pages, impact factor claims, submission management systems). Structural exploitation: the features being exploited, the author's incentive to publish, the evaluator's reliance on journal name recognition, the institutional pressure to produce output, are not removable without dismantling the legitimate incentive architecture of academic publishing. Self-reinforcement: each accepted article creates a bibliographic record that generates further invitations (the author is now a confirmed published academic), fees that fund expanded operations, and, crucially, a published 'journal' that makes the next round of mimicry more convincing.

PSO is distinct from three related frameworks that appear in the predatory publishing literature. It differs from the principal-agent model (Akerlof 1970; Jensen and Meckling 1976) in that PSO does not require information asymmetry between a principal and a delegated agent: the parasite is not an agent of the author-victim and owes no fiduciary duty. It differs from regulatory capture (Stigler 1971) in that no regulatory body is captured: the PSO operates in the interstices between regulatory jurisdictions. It differs from moral hazard frameworks in that no prior contract creates the risk-shifting: the author has not insured against the cost of predatory publication. The distinguishing feature of PSO is its evolutionary mechanism: the parasite is not a defector within a known game, but a replicator that has co-evolved with its host order's defenses.

3. METHODOLOGY

3.1 Corpus Construction

The empirical corpus consists of eight unsolicited journal invitation emails that I personally received between 2025 and early 2026, in the email account associated with my research activity on Zenodo and ResearchGate. This is not a dataset assembled from third-party reports or institutional surveys: each case is a message I received, read, and decided to examine

systematically. The selection criterion was exhaustive within that scope: every unsolicited invitation to submit a manuscript to a journal or conference I had not previously contacted was retained and coded. Eight cases met the criterion. One of them, upon verification, turned out to be a genuine invitation from a recognized open-access publisher with transparent editorial policies. I retained it as a control rather than excluding it, because its presence is analytically useful: it shows what a legitimate invitation looks like against the same coding instrument.

I acknowledge the limitations of this corpus explicitly. Eight cases cannot support statistical inference about the population of predatory publishing operations. The corpus reflects the selective environment of a specific author profile: a researcher publishing in evolutionary legal theory and institutional economics, active on Zenodo, with ORCID registration, receiving invitations in both English and Spanish. Generalizability to other disciplinary niches or geographic contexts requires separate investigation. What the corpus can support, and what I use it for, is the validation of a classification instrument and the reconstruction of a lifecycle model from documented evidence. For a framework paper, this is the appropriate evidential standard; for epidemiological claims, it is not.

Complete documentation of all eight cases, including the original emails with sender metadata, the verification procedures applied, and the coding sheets, is archived in the GitHub repository cited in the title page. The repository is publicly accessible and structured to permit independent replication of the coding and lifecycle reconstruction.

3.2 The PSO Index

The PSO Index operationalizes the three-condition PSO definition through five binary dimensions. A case receives a score of 1 on each dimension if the specified criterion is satisfied and 0 otherwise. The full PSO threshold is a score of 4 or 5; cases scoring 3 or below are classified as ambiguous and require supplementary analysis. The control case scores 0 or 1 across all dimensions, providing an empirical floor.

Table 1. PSO Index: Dimensions, Criteria, and Operationalization

Dimension	Label	Criterion (score = 1 if true)	EPT Correspondence
D1	Credential Mismatch	Journal name/scope does not match author's documented research area	Mimicry: exploits author's name recognition heuristic
D2	Mimetic Apparatus	Journal website reproduces ≥ 3 visual/structural markers of legitimate journals (ISSN, editorial board, impact metric claims)	Extended phenotype: invitation as structural arm of the replicator

D3	Hook Architecture	Invitation includes ≥ 1 personalization element (paper title, institutional affiliation, prior publication cited by name)	Parasitic exploitation: uses host's own data against host
D4	Friction Minimization	Submission process described as expedited (≤ 4 weeks review); fee disclosed only post-acceptance or absent from invitation	Niche stability: reduces exit cost at key decision point
D5	Automation Signature	Email exhibits ≥ 2 of: mass-dispatch headers, generic salutation despite claimed personalization, template syntax artifacts, sender domain mismatch with journal domain	Spontaneous order: operation achieves scale without individual design

The index was developed inductively from the corpus and deductively from the PSO theoretical requirements. I coded all eight cases independently on two occasions separated by four weeks; discordant codes were resolved by returning to the original email text. No case required more than one resolution round. The control case scored $D1=0$, $D2=1$, $D3=1$, $D4=0$, $D5=0$ (total=2), confirming that the index correctly identifies the one documented legitimate invitation as below the PSO threshold. I do not claim this validation is sufficient for generalization; I claim it demonstrates internal consistency.

3.3 Pattern Coding

Beyond the binary PSO Index, I conducted qualitative pattern analysis across the seven PSO-positive cases. A pattern was defined as a structural feature appearing in three or more cases that corresponds to a theoretically identifiable mechanism. I identified six patterns, described in Section 4.2. Pattern coding was conducted using the Boyatzis (1998) protocol for thematic analysis: open coding, axial coding, and selective coding with theoretical saturation as the stopping criterion. Saturation was reached at the sixth pattern, with no new structural features emerging in the seventh case.

4. RESULTS

4.1 Descriptive Statistics

Seven of eight cases (87.5%) satisfied the full PSO threshold (score ≥ 4). The single control case scored 2. Among the seven PSO-positive cases, mean PSO score was 4.57 (SD 0.53), with a

range of 4 to 5. No case scored exactly 3, which is consistent with the theoretical prediction that the PSO threshold corresponds to a structural discontinuity: below it, the operation cannot sustain the mimicry required for niche stability; above it, mimicry is sufficient to generate a positive response rate even among sophisticated targets.

Table 2. Corpus Overview and PSO Index Scores

Case	Journal (anonymized)	D1	D2	D3	D4	D5	Total	Classification
C1	Global Journal of Law and Policy	1	1	1	1	1	5	PSO-Positive
C2	International Review of Legal Science	1	1	1	1	1	5	PSO-Positive
C3	Journal of Jurisprudence and Society	1	1	1	1	0	4	PSO-Positive
C4	World Academy of Legal Research	1	1	0	1	1	4	PSO-Positive
C5	Open Access Law Review (regional)	1	1	1	1	1	5	PSO-Positive
C6	IJLS Special Issue Invitation	1	1	1	0	1	4	PSO-Positive
C7	International Journal of Institutional Theory	1	1	1	1	1	5	PSO-Positive
C8	CTRL [legitimate, control]	0	1	1	0	0	2	Control (legitimate)

Journal names in Table 2 are anonymized consistent with the protocol described in the replication repository. Original metadata, including sender domains, registration dates, and indexing status at time of receipt, is archived there in full.

4.2 Pattern Analysis

Six recurrent structural patterns were identified across the seven PSO-positive cases. Their distribution is presented in Table 3.

Table 3. Pattern Frequency Across PSO-Positive Cases (N=7)

Pattern	Description	PSO cases (n=7)	Control (n=1)	PSO Dimension
P1: Credential mismatch signaling	Journal scope systematically diverges from author's documented specialization	7/7	0/1	D1
P2: Flattery personalization	Invitation names a specific prior publication with praise; phrasing varies across cases but structural template is identical	6/7	1/1	D3
P3: Expedited review promise	Peer review timeline stated as 2-4 weeks; in C3 and C6 the phrase 'fast-track' appears verbatim	6/7	0/1	D4
P4: Fabricated impact metrics	Impact factor or equivalent metric cited without Clarivate/Scimago source; in 4/7 cases the metric format is non-standard	5/7	0/1	D2
P5: Cascading re-invitation	Same or affiliated sender dispatches follow-up within 7-14 days; in C1 three follow-ups were received over 21 days	4/7	0/1	D5
P6: Fee concealment architecture	Article Processing Charge absent from invitation; fee amount discoverable only through journal website navigation of ≥3 clicks	5/7	0/1	D4

Pattern P1, credential mismatch signaling, appeared in all seven predatory cases and in none of the control. This is the PSO framework's most diagnostic signal: a legitimate invitation targets an author because of genuine topical alignment; a parasitic invitation targets an author because the author's name appears in a searchable preprint metadata field, regardless of topic. The mismatch is not a mistake; it is structural evidence of automated scraping. Case C4 provides the starkest illustration: the invitation came from a journal whose stated focus was environmental regulation; the author's indexed work contains no publications on environmental law, a fact verifiable in under sixty seconds by any human editor.

Pattern P2, flattery personalization, appeared in six of seven predatory cases and in the control, which illustrates an important methodological point. Personalization alone is insufficient for PSO classification; its co-occurrence with credential mismatch (P1) is the diagnostic combination. Legitimate editors personalize invitations. Predatory operations also personalize, but the personalization is incongruent with the journal's stated scope. In C2, the invitation named a paper on constitutional lock-in theory and praised its 'contribution to administrative law

scholarship'; neither of those terms appears in the paper, which addresses evolutionary game theory applied to labor reform.

Pattern P6, fee concealment architecture, is theoretically important because it corresponds directly to the friction minimization dimension (D4) of the PSO Index. The Article Processing Charge is the primary resource extracted by the predatory operation, yet it is systematically removed from the point of initial contact. This is not an oversight; it is a design feature. The author who has already drafted a response, looked up the journal, and perhaps submitted a manuscript faces a substantially higher exit cost than the author who sees the fee in the first email. The concealment is phenotypic: it is an adaptation of the replicator that increases the probability of completed extraction.

4.3 PSO Lifecycle Reconstruction

Cross-case analysis of the corpus, combined with metadata from the replication repository, permits reconstruction of the PSO lifecycle in eight stages. I present this as a causal model derived from documented evidence, not as a claim about the internal operations of any specific publisher.

Figure 1 (in replication repository): PSO Lifecycle Diagram

Stage 1: Repository Scraping

Automated agents scan preprint servers (Zenodo, ResearchGate, Academia.edu) and institutional repositories for author metadata. The scraping targets the combination of institutional affiliation, email address, and recent deposit date. In cases C1, C2, and C7, the invitation arrived within 14 days of a new Zenodo deposit, consistent with near-real-time monitoring.

Stage 2: Mimetic Invitation Dispatch

Template-based emails are dispatched with variable personalization inserts drawn from the scraped metadata. The template structure is preserved across operations; what varies is the name of the paper cited and the scope of the target journal. The invitation is the extended phenotype: it is the behavioral arm of the replicator that reaches into the author's decision environment.

Stage 3: Response Filtering

Authors who respond to the invitation are identified as potential extractable targets. The response itself, even a query about the journal's scope, is treated as a signal of tractability. In C5,

a skeptical inquiry elicited a detailed response within two hours, suggesting automated monitoring of reply queues.

Stage 4: Simulated Peer Review

Submitted manuscripts receive 'reviews' within the stated expedited timeline. These reviews are structurally indistinguishable from legitimate reviews in format but are either absent of substantive content or templated. The simulation must be convincing enough to generate the acceptance signal that precedes fee extraction.

Stage 5: Payment Extraction

The acceptance notification triggers the fee disclosure. The amount, typically between USD 200 and USD 1,500 based on published analyses (Shen and Björk 2015; Richtig et al. 2018), is presented as an Article Processing Charge comparable to those of legitimate open-access publishers. The author, having invested time in submission and revision, faces a sunk-cost dynamic.

Stage 6: Credential Vending

Upon payment, the article is assigned a DOI and published. The DOI provides a verifiable record that the author can include in a curriculum vitae. This is the product being sold: not a peer-reviewed publication but a credential that resembles one. The distinction matters because the demand is not for publication per se but for the institutional signals that publication produces.

Stage 7: Author Network Exploitation

Published authors receive invitations to join editorial boards, serve as reviewers, or recommend colleagues for submission. Each of these actions extends the operation's apparent legitimacy. An editorial board member with an institutional affiliation and a verifiable publication record is a phenotypic extension of the replicator into the legitimate academic network.

Stage 8: Niche Stabilization

The operation achieves niche stability when its mimicry is sufficient to evade detection by the countermeasures in operation at a given time. This is not a permanent state: new blacklists, indexing exclusions, or detection tools create new selection pressure. But the lag between countermeasure deployment and operation adaptation has, historically, been favorable to the parasite.

5. DISCUSSION

5.1 Why PSO is Stable: The Selective Environment

The PSO framework's central claim is that predatory publishing persists not despite countermeasures but partly because of them. Beall's lists created selection pressure for improved mimicry: journals delisted for explicit fee-disclosure violations reappeared with better-concealed fee structures. DOI registration requirements, introduced to filter non-indexed publications, were met by wholesale acquisition of legitimate-appearing DOI prefixes. Each countermeasure that targeted a specific phenotypic feature of the predatory operation selected for variants that lacked that feature.

This is the evolutionary dynamic that distinguishes PSO from simple fraud. A fraud scheme cannot adapt in this way because it depends on the deliberate choices of identifiable actors. PSO adapts because the selection operates on variation in a population of operations, most of which are short-lived, without requiring any single actor to design the adaptation. The operations that survive are those that happen to exhibit the phenotypic features the current countermeasures do not target. Their survival generates the template for the next generation.

The practical implication is that countermeasures directed at phenotypic features, blacklists, visual checklists, author education campaigns, will always operate at a disadvantage relative to the adaptive capacity of the PSO. This does not mean they are useless; they raise the cost of entry for new operations and protect some portion of the author population. But they cannot eliminate the niche because they do not alter the selective environment that sustains it.

5.2 Comparison to Competing Frameworks

The principal-agent framework (Jensen and Meckling 1976) analyzes situations in which an agent, delegated to act on behalf of a principal, exploits informational advantages to advance its own interests. Predatory publishing has been analyzed through this lens as a case in which the journal acts as an agent for authors seeking publication, then defects on its obligations. The PSO framework improves on this account in two respects. First, no principal-agent relationship exists in the predatory invitation context: the author has not delegated anything to the journal at the point of receiving an invitation. Second, the principal-agent model predicts that the relationship will be corrected when the principal acquires better information. The persistence of predatory publishing despite widespread author education suggests that informational remedies are insufficient, which the PSO framework explains: the author's decision environment is not merely informationally deficient but structurally exploited.

The regulatory capture framework (Stigler 1971; Peltzman 1976) explains how regulated industries come to control their regulators. It does not apply directly to predatory publishing because there is no regulatory body with jurisdiction over journal quality that the predatory

operation could capture. What the predatory operation captures is not a regulator but an author's cognitive heuristics: the use of journal prestige signals as proxies for peer review quality. The PSO framework describes this capture more precisely as the exploitation of a phenotypic vulnerability in the academic incentive system.

The moral hazard framework applies when one party takes excessive risk because another party bears the cost. It is not the appropriate framework for predatory publishing because no prior contractual relationship creates the risk allocation. What exists is closer to a false advertising situation, but that framing, too, misses the evolutionary dynamic: false advertising is a deliberate strategy, while PSO is an emergent property of a selective process.

5.3 Implications for Institutional Design

If PSO is an emergent equilibrium maintained by selection pressure on the academic incentive architecture, then durable countermeasures must alter that architecture rather than catalogue its symptoms. Three directions follow from the analysis.

First, evaluation systems that weight publication count independently of journal quality create the demand that PSO supplies. Hiring and promotion criteria that treat a publication in a predatory journal identically to no publication would eliminate the instrumental value of credential vending. This is not a novel observation, but the PSO framework gives it theoretical grounding: the credential vending stage (Stage 6) is viable only if the institutional consumer of credentials cannot efficiently distinguish them. Reducing that information asymmetry at the institutional level, through verified publication records, preregistered review protocols, or direct assessment of research quality, alters the selective environment rather than one of its phenotypic manifestations.

Second, the automation signature (D5) of the PSO operation is, paradoxically, its most detectable feature and the least targeted by current countermeasures. Mass-dispatch invitation emails leave identifiable metadata traces: header analysis, sender-domain mismatch detection, and cross-institutional corpus analysis could produce real-time classification of incoming invitations at the institutional email server level. The replication repository associated with this paper includes a prototype classification heuristic based on the D5 criteria, intended as a starting point for such tools.

Third, the spontaneous order character of PSO suggests that coordination among legitimate publishers, indexing services, and institutional repositories could raise the entry cost for new predatory operations by making the mimicry more expensive to execute. If legitimate journals adopt standardized, machine-verifiable identity markers, the marginal cost of producing a convincing mimetic apparatus increases. This is not a solution; it is a measure that shifts the selection equilibrium toward higher-quality mimicry operations, which may or may not improve net author welfare. The trade-off requires empirical study.

5.4 Limitations and Future Research

Several limitations of this study require honest statement. The corpus of eight cases is adequate for instrument development and lifecycle reconstruction but inadequate for prevalence estimation, comparative analysis across disciplines, or testing of the PSO framework's predictions about adaptation dynamics. A large-scale corpus study, ideally with institutional partners willing to share invitation metadata, would be the appropriate next step.

The PSO lifecycle model is reconstructed from observable outputs, the emails themselves and their metadata, without direct access to the operations that produced them. Stage 3 through Stage 7 are inferred from cross-case analysis and existing literature (Shen and Björk 2015; Grudniewicz et al. 2019; Cobey et al. 2018) rather than directly observed. A study with access to the internal processes of a predatory operation, or with data from authors who completed the full lifecycle including payment, would allow more direct testing.

The PSO Index, while internally consistent on this corpus, has not been validated against an independently constructed large-scale dataset. Its five dimensions were derived from a combination of theoretical reasoning and inductive corpus analysis; the weights (all dimensions treated as equally binary) are not empirically calibrated. A confirmatory study with a larger, multi-author corpus would be needed before the index is used for policy purposes.

Finally, the paper focuses on invitation emails directed at researchers in legal theory and institutional economics. Whether the PSO structure identified here generalizes to other disciplinary niches, particularly natural sciences and medicine, where the stakes of predatory publication include patient safety, requires separate investigation.

6. CONCLUSION

This paper has argued that predatory publishing is best understood not as fraud but as an emergent institutional niche: a Parasitic Spontaneous Order that exploits the functional architecture of legitimate academic publishing through a mechanism that EPT identifies as phenotypic extension and Hayek's framework identifies as spontaneous order.

The argument has three layers. Theoretically, the synthesis of EPT and spontaneous order theory produces a framework with predictive content: PSO predicts persistence without conspiracy, adaptation under countermeasure pressure, and selection for improved mimicry over time. Empirically, a five-criterion PSO Index, validated against a corpus of eight documented cases, correctly classifies seven of seven predatory invitations and one of one legitimate control. The lifecycle model reconstructed from corpus evidence identifies eight structurally necessary stages, each of which corresponds to a theoretically predicted function.

The practical upshot is that blacklists and author education, while not useless, address the phenotypic expressions of PSO rather than the selective environment that sustains it. Durable

countermeasures require altering the institutional demand for credentials regardless of source, developing automated detection tools that exploit the automation signatures of PSO operations, and coordinating standardized identity infrastructure among legitimate publishers.

The framework is falsifiable. If predatory publishing operations do not exhibit increasing mimicry quality over time as countermeasure pressure increases; if the PSO Index fails to discriminate predatory from legitimate invitations in a large-scale cross-disciplinary corpus; or if the lifecycle model's predicted stages are absent in direct observation studies of predatory operations, then the PSO account requires revision. These are empirical questions. The replication repository provides the baseline against which future tests can be conducted.

Academic publishing fraud is, in one sense, a narrow institutional problem. In another sense, it is a case study in how parasitic orders emerge and stabilize at the expense of productive institutional architectures. The EPT-Hayek synthesis developed here may be applicable beyond academic publishing: wherever a legitimate order produces verifiable signals of value, a parasitic order has the structural conditions to emerge. Understanding the mechanism is the necessary first step toward effective institutional response.

Acknowledgments

The corpus analyzed in this paper consists of unsolicited invitations received in my own institutional email account between 2025 and early 2026. I am therefore simultaneously the researcher and the subject of study, a position that introduces observer bias I have attempted to control through explicit coding protocols and archiving of raw materials. Computational assistance in document structuring was provided by AI tools; all analytical claims, coding decisions, and interpretations are my own. The author declares no conflicts of interest. This research received no external funding.

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